

RACHEL CHAE

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EDUCATION

Brown University

Bachelor of Science - Computer Science | GPA: 4.0

Providence, RI

Graduating May 2025

WORK EXPERIENCE

Microsoft

Software Engineering Intern (Azure Security)

Remote

Jun - Aug 2024

- **Developed** a Nuclei scanning agent in C#, building a **Docker** script to containerize and deploy the application on Azure Kubernetes Service (AKS), scanning the backend of thousands of Microsoft assets to reduce open vulnerabilities.
- **Architected the solution for seamless scalability and high availability**, ensuring efficient resource management and workload distribution across multiple nodes, **saving Microsoft \$4 million** in just licensing fees from off-the-shelf commercial vendors.

Logitech

Software Engineering Intern (Embedded Audio Firmware Team)

Remote

Jun - Sep 2023

- Creating a cutting-edge application that seamlessly translates **Python** into an **intuitive graphical user interface**, empowering C programmers to effortlessly integrate machine learning algorithms into audio firmware, enhancing usability & accessibility.
- Deploying a standalone engineering software for **digital signal processing** to analyze and build biquad filter designs, significantly accelerating the development process of filter designs and stabilizing audio output amongst speaker products.

IBM Accelerate Program

Software Developer Track

Remote

Jun - Aug 2023

- Practicing Agile and React methodologies through creation of frontend web-applications and gained solid understanding of concepts in cloud native development, application security, & software development lifecycle.
- Participated in weekly group-based projects amongst students and actively led conversations in retrospective meetings to ensure timely and iterative delivery of software projects.

Magna International

Application Engineer Intern

Troy, MI

Jun - Aug 2022

- Developed a robust **General Cybersecurity Encryption Program** in **Python**, utilizing **AES-128 Algorithm** and **Cipher-Block Chain**, to generate CMAC values. This program was expanded by 5 software teams to meet the unique requirements of prestigious clients such as Toyota, Mazda, and General Motors.
- Designed an optimized **Value Stream Map for Toyota's Secure Key Flow and Provisioning Process**, streamlining operations and ensuring maximum security.

LEADERSHIP & PROJECTS

Brown Daily Herald (BDH) Mobile App Team

Frontend Software Developer Team Lead

Providence, RI

Sep 2023 - Present

- Contributed to design and implementation of user-centric features on the BDH's mobile app, like push notifications, authentication, for-you page, in aim of enhancing user engagement through seamless interfaces.

Brown University Department of Computer Science

Teaching Assistant - Cybersecurity & International Relations Masters Course

Providence, RI

Dec 2022 - Present

- Collaborated with a team of 4 teaching assistants and the professor to optimize a graduate-level course, enabling students to acquire **technical cybersecurity expertise** and develop analytical skills for global cybersecurity challenges.

Creative Coding Series

Founder and Project Lead | Brown/RISD STEAM Secretary & Outreach Board Member

Providence, RI

Sep 2021 - Present

- Created the **Creative Coding Workshops** through Brown/RISD STEAM, uniting students from interdisciplinary fields to generate **interactive art exhibitions** through **JavaScript** and hosted **UI/UX workshops** for website development
- **Coding X Fashion** - Led a team of 4 students in the Coding X Fashion project, leveraging creativity, diverse perspectives, and technical expertise to engineer and **code innovative attire** in **Python**.

HONORS & AWARDS

Automatic Coordination of Robotic Teams - Undergraduate Research Assistant

Aug 2023 - Present

- Researched mechanisms and developed web-app alongside Prof. Nora Ayanian to automate a team of ground robots and quadcopters to generate art, integrating both software interfaces and hardware systems.

Automotive Women's Alliance Foundation (AWAF) - 2x Scholarship Recipient | Member

Mar 2021 - Present

- Involved in promoting the advancement of women engineers and closing the age gap of female leaders in the automotive industry.

SKILLS & COURSES

Relevant Coursework: Computer Systems, Software Engineering, Deep Learning, Machine Learning, Computer Vision, UI/UX, Probability & Statistical Inference, Discrete Mathematics, Linear Algebra, Software Security & Exploitation

Technical Skills: Python, Java, C, HTML, CSS, Figma, JavaScript, TypeScript, React, NodeJS, Unix/Linux, Shell, Git, Jira, Excel